

Paper: [Early Detection of Windows Cryptographic Ransomware Based on Pre-Attack API Calls Features and Machine Learning](#)

Group ID : 47

1. Overview and Problem Statement The paper by Zakaria et al. (2024) addresses the escalating threat of cryptographic ransomware, which utilizes crypto-viral extortion techniques to encrypt victim data and demand payment. Traditional detection methods typically focus on the encryption or post-attack stages. The authors argue that detecting ransomware during mass file encryption is often too late, resulting in irreversible data loss. The central problem is the lack of visible early behaviour signs, making it challenging to stop the malware before it acts.

2. The RENTAKA Framework To solve this, the authors proposed RENTAKA, a novel machine learning framework designed to detect cryptographic ransomware during the pre-encryption phase. The framework is built on an understanding of the ransomware lifecycle, which consists of five phases: Deployment, Installation, Command & Control, Destruction (Encryption), and Extortion. RENTAKA specifically targets the "Setup" phase (encompassing deployment, installation, and C2).

By executing dynamic analysis in a Cuckoo Sandbox, the algorithm extracts a sequence of Windows API calls. It then searches the behavioural log for an encryption API marker. Once found, the algorithm isolates only the API calls that occurred *before* the encryption flag, generating a purely pre-encryption dataset.

3. Methodology and Findings The researchers utilized a dataset from the RISS research group containing 582 ransomware samples across 10 families and 942 benign software samples. They tested five standard classifiers: Random Forest, Naive Bayes, Support Vector Machines (SVM), k-Nearest Neighbors (kNN), and J48.

The paper concluded that the SVM model yielded the best overall performance. According to their results, SVM achieved a 97.0492% accuracy rate with a high True Positive Rate of 0.995 and a low False Positive Rate of 0.071. Naive Bayes performed the worst, logging an accuracy of only 80.9836%.

4. Critique and Real-World Applicability The RENTAKA methodology is highly effective conceptually. By filtering out post-encryption noise, it forces the machine learning models to identify the subtle setup behaviors of ransomware (such as payload persistence or environment mapping).

However, a critical limitation of the paper's experiment is its dataset distribution. The authors used 582 ransomware samples against 942 benign samples, representing a roughly 38% infection rate. In a real-world enterprise environment, ransomware events represent a tiny fraction of total network and API activity. As demonstrated by our own implementation run on the 14,600+ sample VirusShare dataset, introducing a heavy class imbalance causes standard models to struggle with recall (TPR), even if overall accuracy remains high. Future research

should test the RENTAKA feature-set against highly imbalanced datasets to better simulate zero-day enterprise detection.

2301AI46 : Kumar Anurag

2301AI45 : Ayush Kumar Singh